EDGEERGO — UI/UX DESIGN DOCUMENT

Product: EdgeErgo
Document Type: Product UI/UX Design Specification
Design Direction: Premium Dark Enterprise Analytics Dashboard
Version: 1.0
Platform: Desktop Web Dashboard + Worker Desktop Overlay
Primary Users: Administrators, Supervisors, Ergonomics/HR Teams, Workers

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1. DESIGN OVERVIEW

EdgeErgo is a privacy-preserving ergonomic assessment platform that uses edge-based computer vision to analyze workplace posture, movement, fatigue-related indicators, and ergonomic risk.

The interface must communicate:

- Professionalism
- Privacy
- Reliability
- AI-powered analytics
- Workplace safety
- Real-time monitoring
- Data-driven decision making

The visual language is inspired by modern financial analytics dashboards:

- Dark interface
- Minimal visual noise
- Rounded analytical cards
- High information density
- Strong typography hierarchy
- Thin borders
- Subtle status colors
- Interactive charts
- Compact navigation
- Large KPI values

The interface should **not look like a medical application or fitness tracker**.

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2. DESIGN PRINCIPLES

2.1 Minimal

Every element should have a purpose.

Avoid unnecessary:

- Illustrations
- Decorative gradients
- Large icons
- Excessive animations
- Excessive colors

2.2 Data First

The dashboard should prioritize measurable ergonomic information.

Important information should be visible without navigating through multiple pages.

Examples:

- Workers monitored
- Posture compliance
- RULA score
- REBA score
- High-risk events
- Active alerts
- Session duration
- Break adherence

2.3 Privacy First

Privacy must be visible within the interface.

The system should clearly communicate:

Processing occurs locally on the worker device.

The dashboard should never imply that administrators are viewing live camera footage.

2.4 Calm Alerts

Alerts should inform rather than alarm.

Use:

- Green → Safe
- Yellow → Attention
- Red → High Risk

Avoid flashing interfaces or aggressive warning animations.

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3. VISUAL DESIGN SYSTEM

3.1 Color Palette

Primary Background

Application Background
#000000

Page Background

#39383D

Cards

Primary Surface
#1D1D1D

Secondary Surface
#222222

Hover Surface
#2B2B2B

Text

Primary
#F5F5F5

Secondary
#A7A7A7

Muted
#707070

Disabled
#505050

Brand

Primary Brand
#D98CB3

Secondary Brand
#A878FF

Data Accent
#66C7FF

Status

Safe
#39D98A

Attention
#F2C94C

High Risk
#FF6B7A

Critical
#FF4D4D

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4. TYPOGRAPHY

Font

Inter

Fallback:

system-ui,
-apple-system,
BlinkMacSystemFont,
"Segoe UI",
sans-serif

Typography Scale

Element	Size	Weight
Display Metric	30–32px	500
Page Title	26px	500
Section Heading	18px	500
Card Heading	14px	500
Body	13px	400
Caption	11px	400
Navigation	12–13px	400/500

Typography should remain compact.

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5. SPACING SYSTEM

Use an 8px spacing system.

4px
8px
12px
16px
24px
32px
40px

Recommended:

Card padding: 16–20px
Card gap: 8–12px
Section gap: 24px
Navigation gap: 20–28px

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6. BORDER AND RADIUS SYSTEM

Small radius: 6px
Medium radius: 10px
Card radius: 14px
Large radius: 16px
Pill radius: 999px

Borders:

#303030

Use extremely subtle borders rather than heavy shadows.

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7. APPLICATION STRUCTURE

The EdgeErgo application consists of:

EdgeErgo
│
├── Dashboard
├── Workers
├── Devices
├── Analytics
├── Reports
└── Alerts

Top-right:

System Status
Notifications
User Account

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8. GLOBAL NAVIGATION

The top navigation should contain the EdgeErgo logo on the left.

EDGEERGO

Dashboard
Workers
Devices
Analytics
Reports
Alerts

Right side:

● System Online
🔔
User

The active navigation item should use a slightly brighter background.

Example:

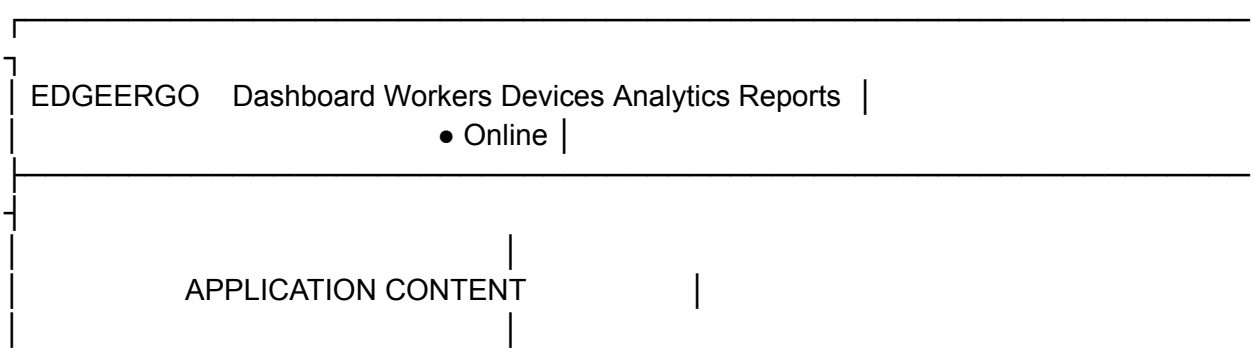
Dashboard
[redacted]

instead of an aggressive colored highlight.

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9. APPLICATION SHELL

Desktop layout:



└──────────────

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12. KPI SECTION

The primary dashboard should contain four important KPIs.

KPI 1 — Workers Monitored

Workers monitored

24

+3 today

KPI 2 — At-Risk Workers

At-risk workers

05

12.4% change

KPI 3 — Posture Compliance

Posture compliance

82.6%

+4.8%

KPI 4 — Active Alerts

Active alerts

07

3 high priority

Each KPI should be presented inside a dark rounded card.

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13. POSTURE RISK TIMELINE

Primary dashboard visualization:

Posture Risk Timeline

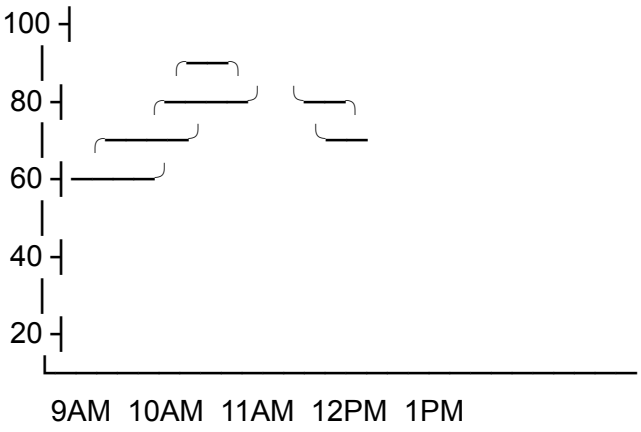
The chart displays ergonomic changes over time.

Possible metrics:

RULA
REBA
Posture %
Fatigue Indicator

User can switch between metrics.

Example:



The graph should use subtle grid lines.

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14. POSTURE OVERVIEW CARD

Card title:

Posture Overview

Primary metric:

82.6%

Subtitle:

Overall posture compliance

Breakdown:

Good 82.6%
Moderate risk 11.8%
High risk 5.6%

Visual representation:



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15. RISK ANALYSIS CARD

Card title:

Risk Analysis

Use a donut chart.

Center:

130
Assessments

Categories:

Low Risk
58%

Moderate Risk
27%

High Risk
11%

Critical
4%

Hovering over a section should display:

High Risk
14 assessments
11%

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16. ERGONOMIC RISK FORECAST

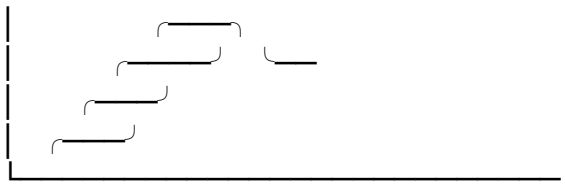
Card title:

Ergonomic Risk Forecast

The chart displays historical and estimated risk patterns.

Example:

Risk



Mon Tue Wed Thu Fri

Forecast information must clearly be labelled as an estimate.

Example:

Expected elevated-risk period

Tomorrow
2:00 PM – 4:00 PM

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17. INSIGHT CARD

At the bottom of the dashboard:

✦ Insight

5 workers have maintained elevated-risk posture

for more than 20 minutes.

Consider a short movement break.

The insight card should use a neutral dark background.

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18. WORKERS PAGE

The Workers page displays all monitored workers.

Workers

24 active workers

[Search workers...] [Filter ▼]

Table:

Worker	Device	Status	RULA	REBA	Risk	Session
Worker 01	PC-001	Online	3	4	Low	2h 15m
Worker 02	PC-002	Online	5	6	Moderate	3h 04m
Worker 03	PC-003	Offline	—	—	—	—
Worker 04	PC-004	Online	7	8	High	4h 12m

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19. WORKER DETAIL PAGE

Header:

← Workers

Worker #W024
● Online

Summary cards:

Posture Compliance
82%

RULA
4

REBA
5

Session
5h 24m

---

20. LIVE POSTURE PANEL

Large card:

Live Posture

Display the MediaPipe skeleton.

```
    O
   /|\
   / \
```

Skeleton status:

Green = Good
Yellow = Attention
Red   = High Risk

Display:

RULA 4
Moderate Risk

The actual camera feed should not be shown on the central administrator dashboard.

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21. POSTURE BREAKDOWN

Display individual body regions.

Posture Breakdown

Neck	Moderate
Shoulder	Good
Elbow	Good
Wrist	Moderate
Back	High
Legs	Good

Use status badges.

Example:

- Good
- Moderate
- High

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22. FATIGUE ANALYTICS

Card:

Fatigue & Activity

Metrics:

Continuous screen time
03h 42m

Movement frequency
Low

Break adherence
68%

Posture deterioration
+14%

The system should focus on observable activity metrics rather than making medical diagnoses.

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23. EXPRESSION ANALYTICS

Card:

Observed Expression Patterns

Example:

Neutral 72%
Fatigue-related 18%
Other 10%

Avoid statements such as:

"Worker is stressed."
"Worker is depressed."

The system should describe detected patterns rather than infer psychological conditions.

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24. DEVICES PAGE

The Devices page monitors EdgeErgo worker agents.

Devices

24 registered
21 online
2 warning
1 offline

Table:

Device	Worker	Status	Last Sync	Agent
PC-001	Worker 01	Online	2 sec	v1.0
PC-002	Worker 02	Online	4 sec	v1.0
PC-003	Worker 03	Offline	18 min	v1.0
PC-004	Worker 04	Warning	7 sec	v1.0

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25. DEVICE STATUS

Statuses:

- Online
- Warning
- Offline

Online:

#39D98A

Warning:

#F2C94C

Offline:

#FF6B7A

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26. ANALYTICS PAGE

The Analytics page provides deeper analysis.

Sections:

Posture Trends
Risk Distribution
RULA Analysis
REBA Analysis
Fatigue Trends
Activity Trends
Expression Patterns

Filters:

Worker
Department
Device
Date
Time
Risk Level

---

27. REPORTS PAGE

Header:

Ergonomic Assessment Reports

Report summary:

Workers analyzed 24
Sessions 148
Average RULA 3.8
Average REBA 4.2
High-risk events 17

Charts:

Risk Distribution
Posture Trend
Worker Comparison
Time-Based Risk

Actions:

Export CSV
Generate PDF

---

28. ALERTS PAGE

Alerts should be organized by severity.

All
Attention
High Risk
Critical
Resolved

Alert card:

HIGH RISK

Worker #W024
Elevated posture risk detected.

Duration
24 minutes

RULA
7

12:42 PM

[View Worker]

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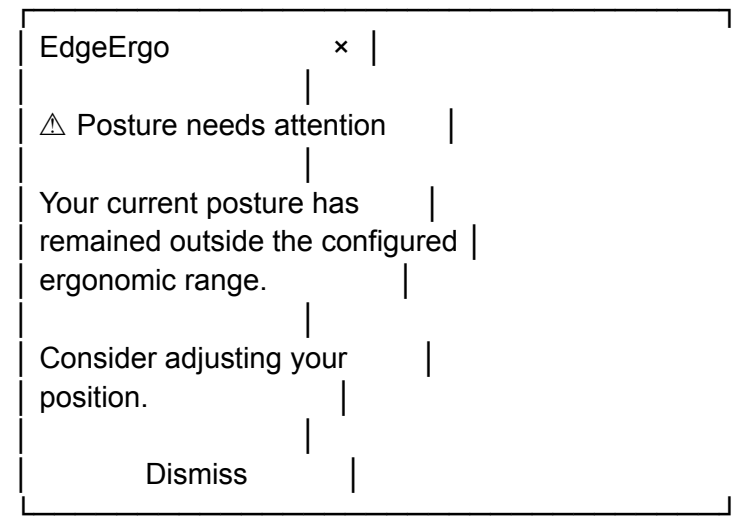
29. WORKER DESKTOP OVERLAY

The worker overlay is separate from the administrator dashboard.

It must be:

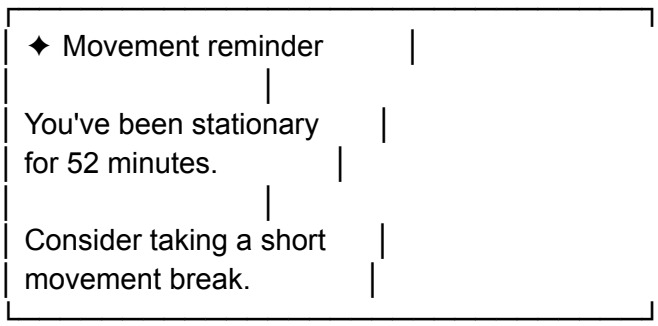
- Small
- Non-intrusive
- Transparent-looking
- Easily dismissible
- Positioned near the screen edge

Example:



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30. MOVEMENT REMINDER



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31. PRIVACY INDICATOR

The worker application should display:

● Local Processing Active

Expanded state:

Privacy Status

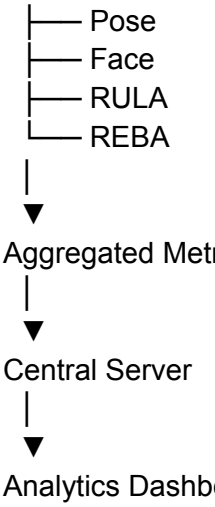
✓ Camera processing is local
✓ Raw video is not uploaded
✓ Pose data is processed locally
✓ Facial analysis is processed locally
✓ Only aggregated metrics are synchronized

---

32. PRIVACY ARCHITECTURE VISUALIZATION

The dashboard can include a privacy information panel:

Camera
  │
  ▼
Local Edge AI
  │



Raw video must not be represented as being uploaded to the central server.

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33. DATA MODEL FOR DASHBOARD

Example data sent by each worker agent:


```
{
  "worker_id": "W024",
  "device_id": "PC-024",
  "timestamp": "2026-09-19T12:15:00",
  "rula_score": 4,
  "reba_score": 5,
  "posture_status": "moderate",
  "sitting_duration": 3240,
  "movement_frequency": 0.42,
  "fatigue_indicator": 0.31,
  "expression_pattern": "neutral",
  "break_adherence": 0.68
}
```


The central dashboard converts this data into:

- KPI cards
- Graphs
- Risk charts
- Worker tables
- Reports

- Alerts

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34. RESPONSIVE DESIGN

Desktop

Primary target:

1440px+

Layout:

Main Analytics KPI Panel
2/3 1/3

Laptop

1024px – 1439px

Reduce:

- Card padding
- Chart height
- Navigation spacing

Tablet

768px – 1023px

Stack dashboard sections vertically.

Mobile

Mobile is not the primary EdgeErgo environment but the dashboard should remain readable.

Cards should become:

1 column

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35. COMPONENT DESIGN

Required reusable components:

AppShell
TopNavigation
PageHeader
Card
KPI Card
StatusBadge
Dropdown
Button
Tooltip
Modal
ChartContainer
WorkerTable
DeviceTable
PostureSkeleton
RiskIndicator
AlertCard
InsightCard
PrivacyIndicator

---

36. COMPONENT STYLE

Cards

Background: #1D1D1D
Border: #303030
Radius: 14px
Padding: 16–20px

No strong shadows.

Buttons

Primary:

Background: #F5F5F5
Text: #000000
Radius: 8px

Secondary:

Background: #252525
Text: #F5F5F5
Border: #303030

Status Badges

Safe
Moderate
High
Critical

Use subtle tinted backgrounds rather than solid bright backgrounds.

---

37. CHART DESIGN

Charts must remain visually consistent.

Chart rules

- Minimal grid lines
- No 3D charts
- No excessive colors
- Rounded tooltips
- Smooth lines
- Clear axis labels
- Interactive hover states

Recommended charts:

Line Chart
Bar Chart
Donut Chart
Area Chart

Timeline

Avoid:

3D pie charts
Radar charts everywhere
Excessive animations

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38. ICONOGRAPHY

Use a consistent outline icon library.

Recommended style:

Lucide-style icons

Icons should generally be:

16px
18px
20px

Avoid oversized icons.

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39. ANIMATION

Animations should be subtle.

Recommended:

150–300ms
ease-out

Use animation for:

- Card hover
- Dropdown
- Tooltip
- Chart transitions
- Alert appearance

Do not animate continuously unless necessary.

---

40. EMPTY STATES

Example:

No workers available

Worker devices will appear here
once EdgeErgo agents connect.

[Add Device]

---

41. OFFLINE STATE

Since EdgeErgo is designed for edge processing:

Central server unavailable

Local monitoring continues on connected
worker devices.

Last synchronization:
2 minutes ago

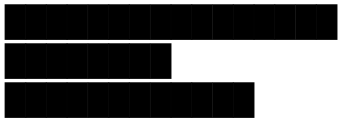
This is an important UX requirement.

---

42. LOADING STATE

Use skeleton loading rather than spinners everywhere.

Example:



Charts should have skeleton placeholders.

---

43. ERROR STATE

Example:

Unable to load worker analytics

The dashboard could not retrieve
the latest aggregated data.

[Retry]

Do not expose raw backend errors to users.

---

44. ACCESSIBILITY

The UI must support:

- Keyboard navigation
- Visible focus states
- Sufficient text contrast
- Semantic HTML
- Screen-reader labels
- Tooltips for unfamiliar icons
- Non-color-only status indicators

For example:

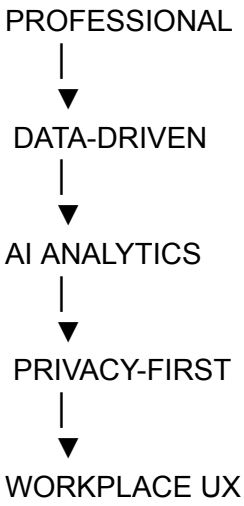
● High Risk

rather than relying only on red color.

---

45. DESIGN LANGUAGE SUMMARY

The final EdgeErgo interface should feel:

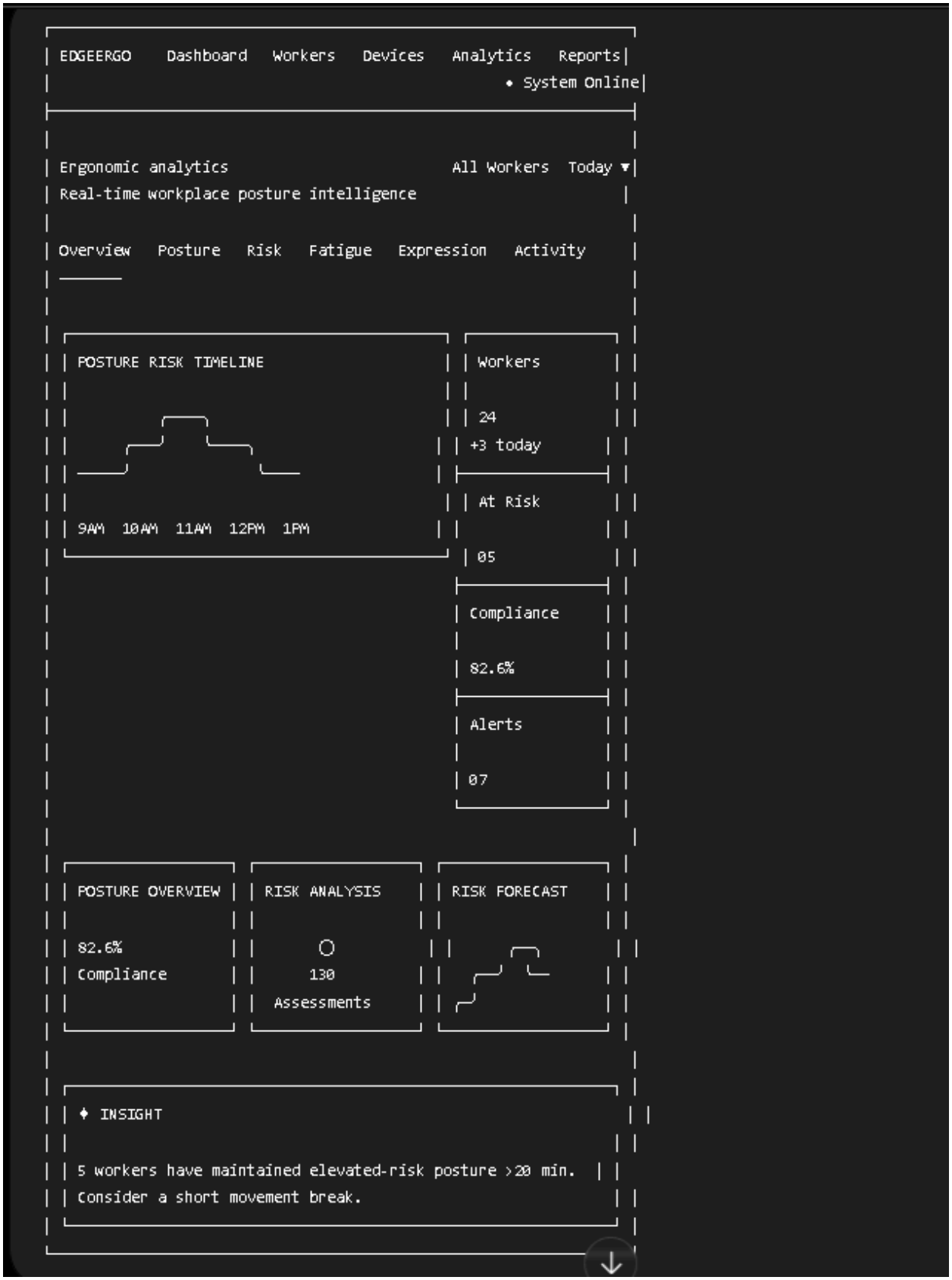


The visual identity should be:

Dark
Minimal
Premium
Analytical
Technical
Calm
Professional
Privacy-focused

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46. FINAL DASHBOARD WIREFRAME


EDGEERGO
Dashboard
Workers
Devices
Analytics
Reports
• System Online
Ergonomic analytics
Real-time workplace posture intelligence
All Workers
Today
Overview
Posture
Risk
Fatigue
Expression
Activity
POSTURE RISK TIMELINE
9AM
10AM
11AM
12PM
1PM
Workers
24
+3 today
At Risk
05
Compliance
82.6%
Alerts
07
POSTURE OVERVIEW
82.6%
Compliance
RISK ANALYSIS
130
Assessments
RISK FORECAST
INSIGHT
5 workers have maintained elevated-risk posture >20 min.
Consider a short movement break.

47. DESIGN GOAL

The final EdgeErgo product should look like a **real enterprise SaaS product**, not a college-project dashboard.

The visual system should consistently follow:

Dark premium interface → compact navigation → rounded analytical cards → large KPIs → clean charts → subtle status colors → privacy indicators → minimal interaction friction.